EARTHWISE AI

The Intelligent Earth Trend Observatory

FORGE CYBER LAB

COMPLETE PROJECT BACKUP & CONTINUATION HANDOFF

Prepared: 11 November 2026

Mission: Understand Earth. Detect Change. Explain the Future.

This document preserves the project vision, current technical status, repository details, known limitations, and the exact next steps so work can resume without repeating setup.

1. Project identity and mission

Product: EARTHWISE AI — The Intelligent Earth Trend Observatory

Organization: FORGE CYBER LAB

Tagline: Understand Earth. Detect Change. Explain the Future.

Build an evidence-based environmental intelligence platform using NASA Earth science data. The platform should identify what is changing, where it is changing, how much it is changing, and whether the observed trend is statistically significant.

Initial geographic focus: Pakistan, with a design that can scale globally. The project is aligned with the NASA Space Apps Challenge theme “Be An Earth System Trend Detective!”

2. Core product concept

- **Living Earth Explorer:** interactive map with trustworthy NASA environmental layers.
- **Hidden Trend Detective:** calculate environmental trends and report statistical evidence.
- **Opposite Earth:** compare two locations using the same variable and time period.
- **AI Earth Scientist:** explain measured results in plain language, including uncertainty and limitations.
- **Environmental Risk Signals:** highlight heat, drought, and heavy-rain indicators without presenting them as guaranteed disaster predictions.

Signature output — Earth Trend Fingerprint™

- Temperature direction: rising / falling / no clear trend.
- Rainfall direction: rising / falling / no clear trend.
- Vegetation direction: improving / declining / no clear trend.
- Statistical status: significant / not significant / inconclusive.
- Data quality: good / limited / insufficient.

3. MVP scope and pilot locations

Pilot cities: Lahore, Islamabad, and Karachi. Candidate analysis period: 2001–2025, only where the chosen dataset has adequate coverage and quality.

Initial variables: air temperature, precipitation, vegetation, and soil moisture. Keep the first working version narrow and scientifically honest; do not attempt a full 3D digital twin in the MVP.

- NASA POWER: temperature and precipitation first.
- NASA GPM IMERG V07: precipitation enhancement after the core app works.
- MODIS MOD13Q1 V061: NDVI/EVI vegetation analysis after QA filtering.
- NASA FIRMS: optional fire detections for context only.

4. Data sources and science notes

NASA POWER

Priority parameters include T2M, T2M_MAX, T2M_MIN, PRECTOTCORR, and RH2M. The existing MVP requests daily point data for temperature and precipitation, using UTC. Confirm API parameter availability, spatial support, missing values, and units before interpreting results.

GPM IMERG V07

Approximate spatial resolution is 0.1° (about 10 km) with 30-minute base observations; the record extends back to 1998. Product latency and version must be documented.

MODIS MOD13Q1 V061

250 m, 16-day NDVI/EVI product. Apply quality assurance flags and the 0.0001 scale factor before analysis.

Statistical method

The current MVP aggregates daily values to monthly observations, then applies Kendall tau and Theil–Sen slope estimation. Report slope units, p-value, confidence interval where available, sample size, and data-quality notes.

Scientific caution: the current method does not correct for monthly seasonality, serial autocorrelation, or multiple hypothesis testing. Do not describe it as publication-grade or make strong causal claims. Future improvements: seasonal Mann–Kendall, modified Mann–Kendall/autocorrelation handling, and false-discovery-rate correction.

5. Current codebase and functionality

A starter project ZIP was previously prepared with the following structure:

```
README.md
requirements.txt
src/nasa_power.py
src/regions.py
src/cleaning.py
src/statistics.py
src/comparisons.py
src/ai_engine.py
app/main.py
tests/test_core.py
docs/methodology.md
docs/data_sources.md
docs/roadmap.md
docs/research_brief.md
.gitignore
data/ (placeholders)
```

- NASA POWER daily point-data retrieval for Lahore, Islamabad, and Karachi.
- Temperature and precipitation as the initial variables.
- Monthly aggregation: temperature mean and precipitation sum.
- Simple Kendall tau and Theil–Sen trend estimates; minimum 24 monthly observations.
- Evidence-constrained deterministic explanation (not a generative AI model).
- Basic Streamlit interface with location/variable selection, data fetch, chart, and statistics.

Not yet implemented: GPM and MODIS data pipelines; advanced statistical corrections; full interactive map; production AI; automated alerts.

Verification status: only a synthetic trend calculation was privately checked. Full pytest suite and live app execution were not confirmed. No live NASA data was fetched in the prior environment. Therefore, no real measured city trend values should be claimed yet.

6. GitHub repository and upload status

Account: Abbubakar-khan

Repository: https://github.com/Abbubakar-khan/earthwise-ai

Git remote: https://github.com/Abbubakar-khan/earthwise-ai.git

Default branch: main | **Visibility:** public

The repository was verified to exist, but it was empty at the last check. The GitHub connector could read the repository but could not write files (403: integration lacks content-write permission). No upload was completed by the assistant.

Manual upload procedure (mobile):

- Download the previously prepared project ZIP from the chat and extract it on the phone.
- Open the repository link above in a browser and sign in.
- Tap Add file → Upload files.
- Select the extracted project files and folders. Upload the contents, not the ZIP archive itself.
- Commit message: Add EARTHWISE AI MVP.
- Tap Commit changes and wait for GitHub to finish uploading.
- Then tell ChatGPT “upload complete” or share a screenshot. The repository contents must be fetched and verified before calling the upload successful.

7. Immediate next work — resume checklist

- **First:** check whether the project ZIP has been extracted and whether the upload screen is open.
- **Second:** finish the manual GitHub upload and commit.
- **Third:** verify the repository file tree and README online.
- **Fourth:** inspect requirements and source files for missing paths/import errors.
- **Fifth:** run the app and tests in a suitable Python environment.
- **Sixth:** fetch one small real NASA POWER sample for one city and verify units, dates, missingness, and aggregation.
- **Seventh:** add clear “demo / preliminary” labels until scientific validation is complete.

8. Product rules — do not compromise

- Never invent NASA measurements, trend values, citations, or live results.
- Never claim a forecast or disaster warning is guaranteed.
- Show sample size, period, units, p-value, slope, and data-quality status.
- Separate correlation/trend detection from causal explanation.
- Keep AI explanations grounded in computed evidence and state uncertainty.
- Document dataset versions, access dates, preprocessing, and limitations.
- Prioritize a small, working, scientifically defensible MVP over a large unfinished feature list.

9. Continuation prompt — paste this in the next chat

> Continue my EARTHWISE AI project from this handoff. I am working alone on mobile and my internet may be slow. The GitHub repository is https://github.com/Abbubakar-khan/earthwise-ai and was last verified as empty; the GitHub connector had read access but no write permission, so I must upload the extracted project ZIP contents manually. Guide me one small step at a time. When I say upload is complete, inspect the actual GitHub repository contents before confirming. Do not invent test results or NASA data. Help me validate the MVP and build the NASA Space Apps Challenge deliverables.

End of backup. Keep this PDF with the project ZIP and use the continuation prompt whenever restarting the work.